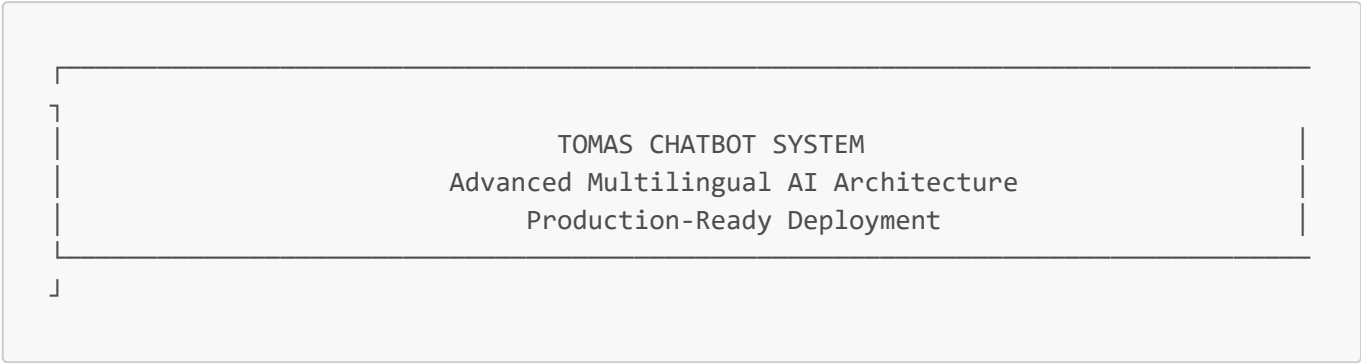
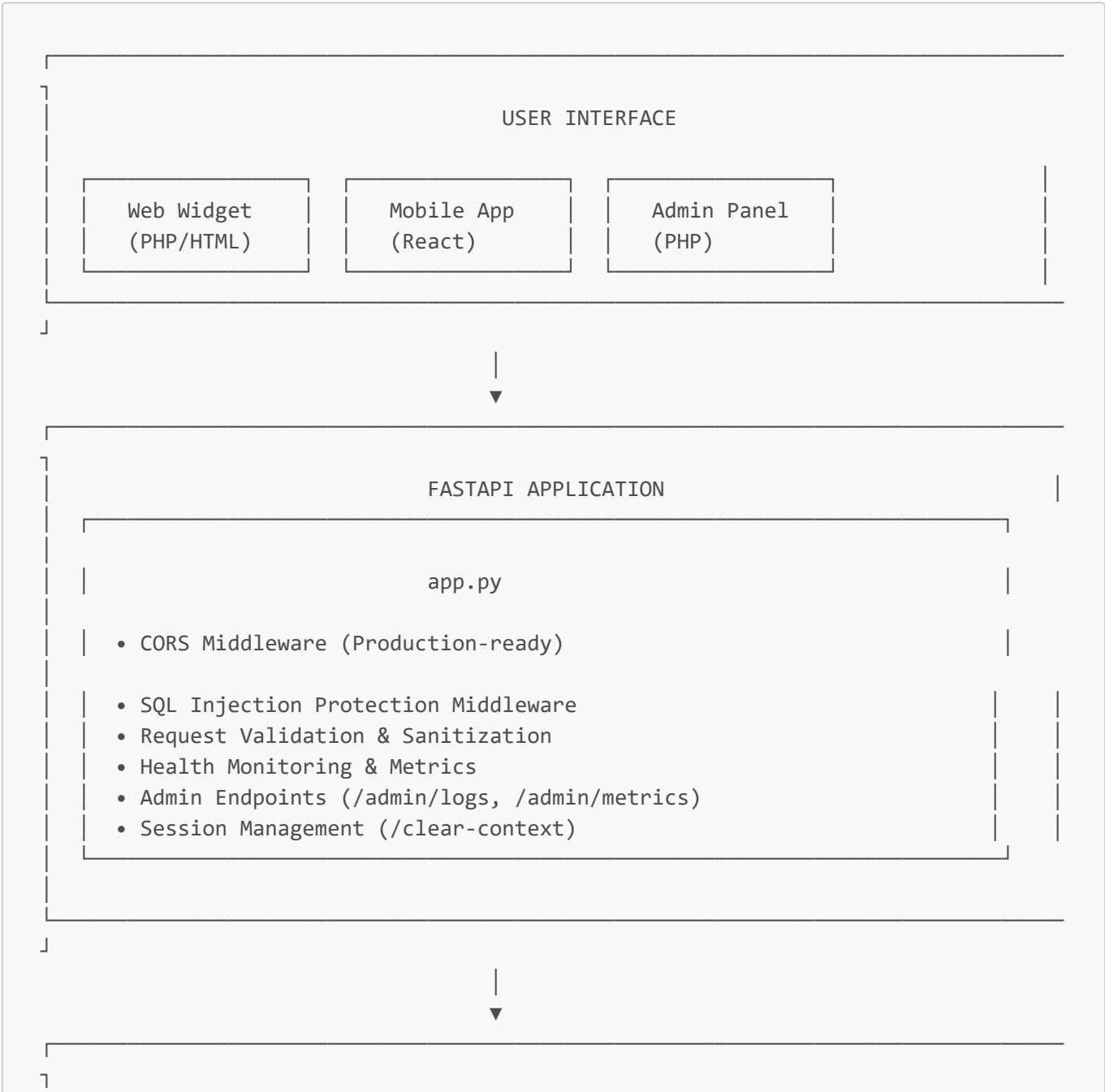


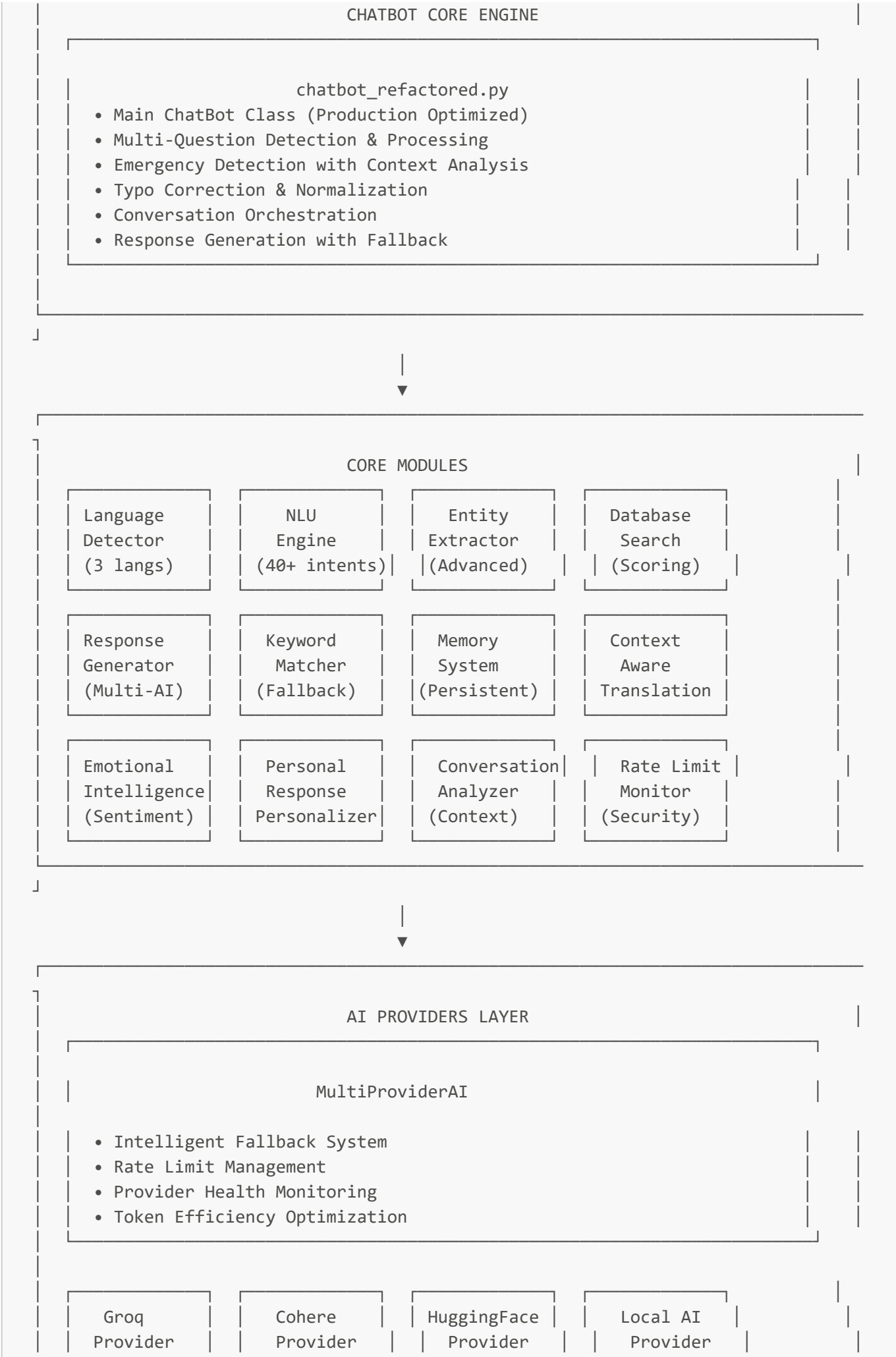
TOMAS Chatbot Architecture Diagram (Updated)

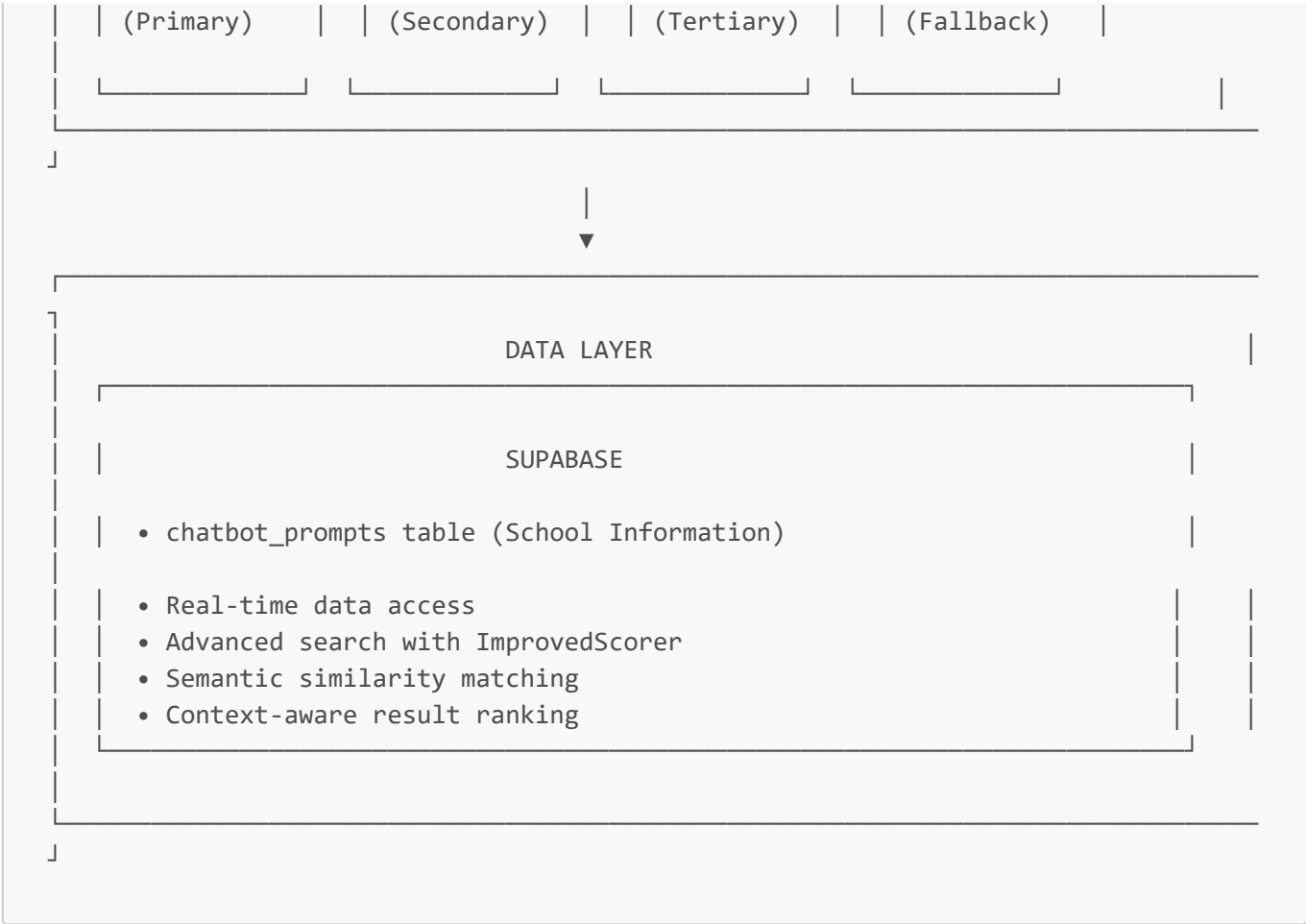
System Overview



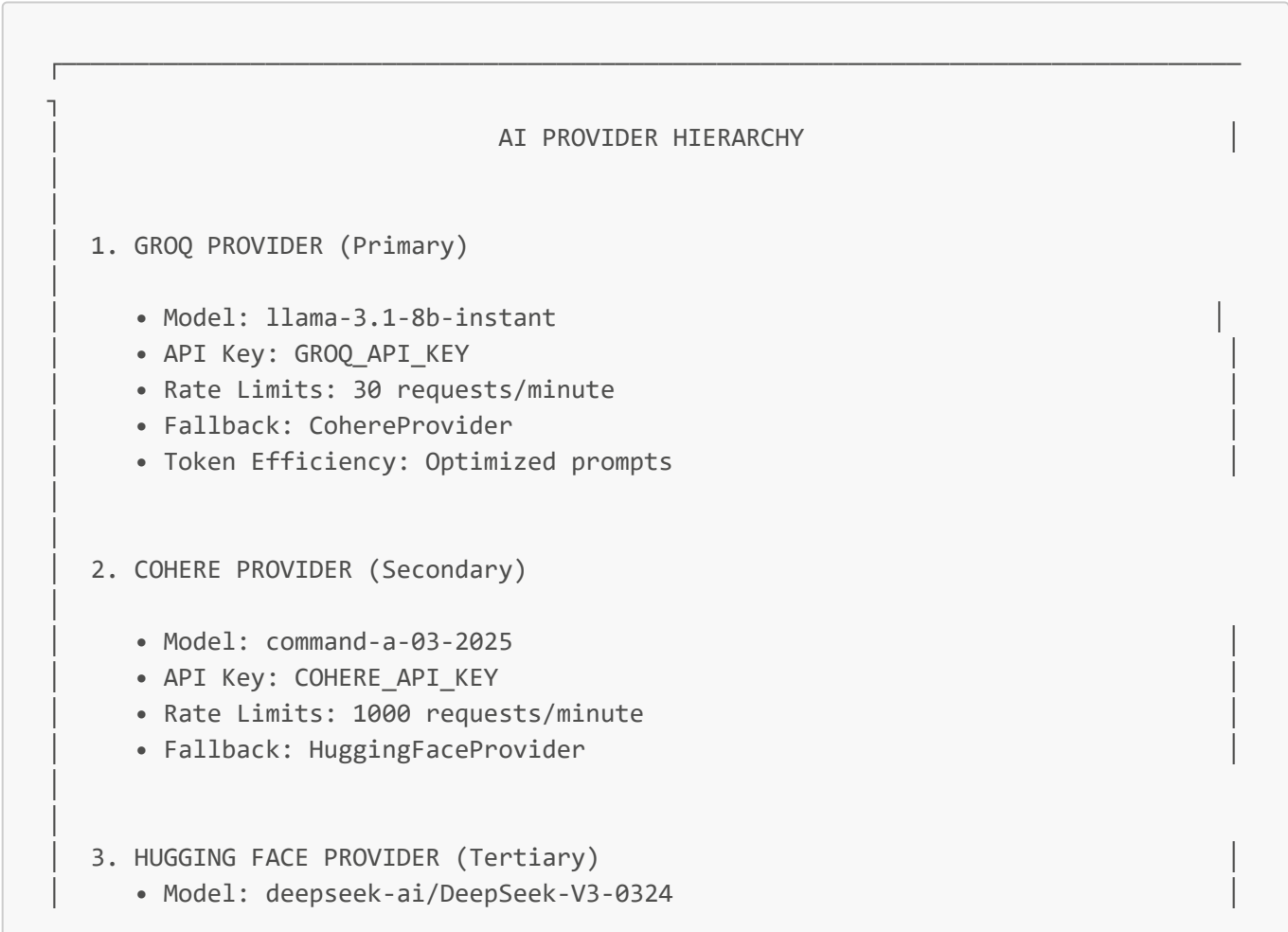
Core Architecture







AI Provider Configuration (Updated)



- API Key: HUGGINGFACE_API_KEY (optional)
| Rate Limits: 1000 requests/day
 - Fallback: LocalAIProvider
4. LOCAL AI PROVIDER (Final Fallback)
- Model: deepseek-ai/DeepSeek-V3-0324
 - API Key: HUGGINGFACE_API_KEY (optional)
 - Rate Limits: Unlimited (free)
 - Fallback: Context-aware keyword responses

Core Features & Capabilities (Updated)

CHATBOT FEATURES	
 MULTILINGUAL SUPPORT	
• English (Primary) • Tagalog (Secondary) • Aklanon (Tertiary) • Mixed-language detection with confidence scoring • Context-aware translation with grammar preservation • Language-specific response formatting	
 INTELLIGENT CONVERSATION	
• Natural Language Understanding (NLU) with 40+ intents • Advanced Entity Extraction (Names, Grades, Subjects, Relationships) • Conversation Memory with persistent user profiles • Context Awareness across multi-turn conversations • Multi-question detection and structured processing • Emergency detection with context analysis	
 SCHOOL-SPECIFIC FEATURES	
• Enrollment Information (Documents, Deadlines, Process) • Staff Directory with role-based queries • Schedule Inquiries (Hours, Classes, Events) • Location Services (Address, Directions, Facilities) • Financial Information (Tuition, Fees, Scholarships) • Facilities Information (CR, Library, Cafeteria, etc.) • Grade Level Validation and Information	

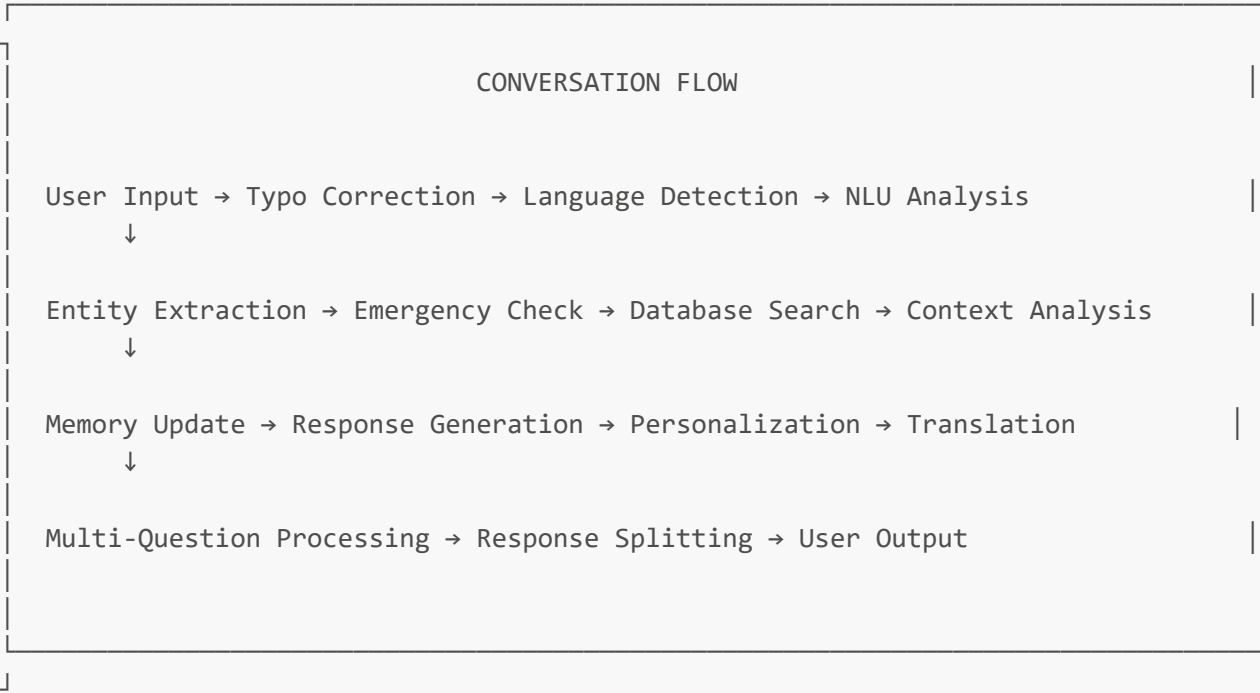
 SAFETY & SECURITY

- Medical Emergency Detection with 911 protocols
- SQL Injection Protection Middleware
- Rate Limiting and Abuse Prevention
- Input Validation and Sanitization
- Gibberish Detection with NLP patterns
- Context-aware emergency analysis

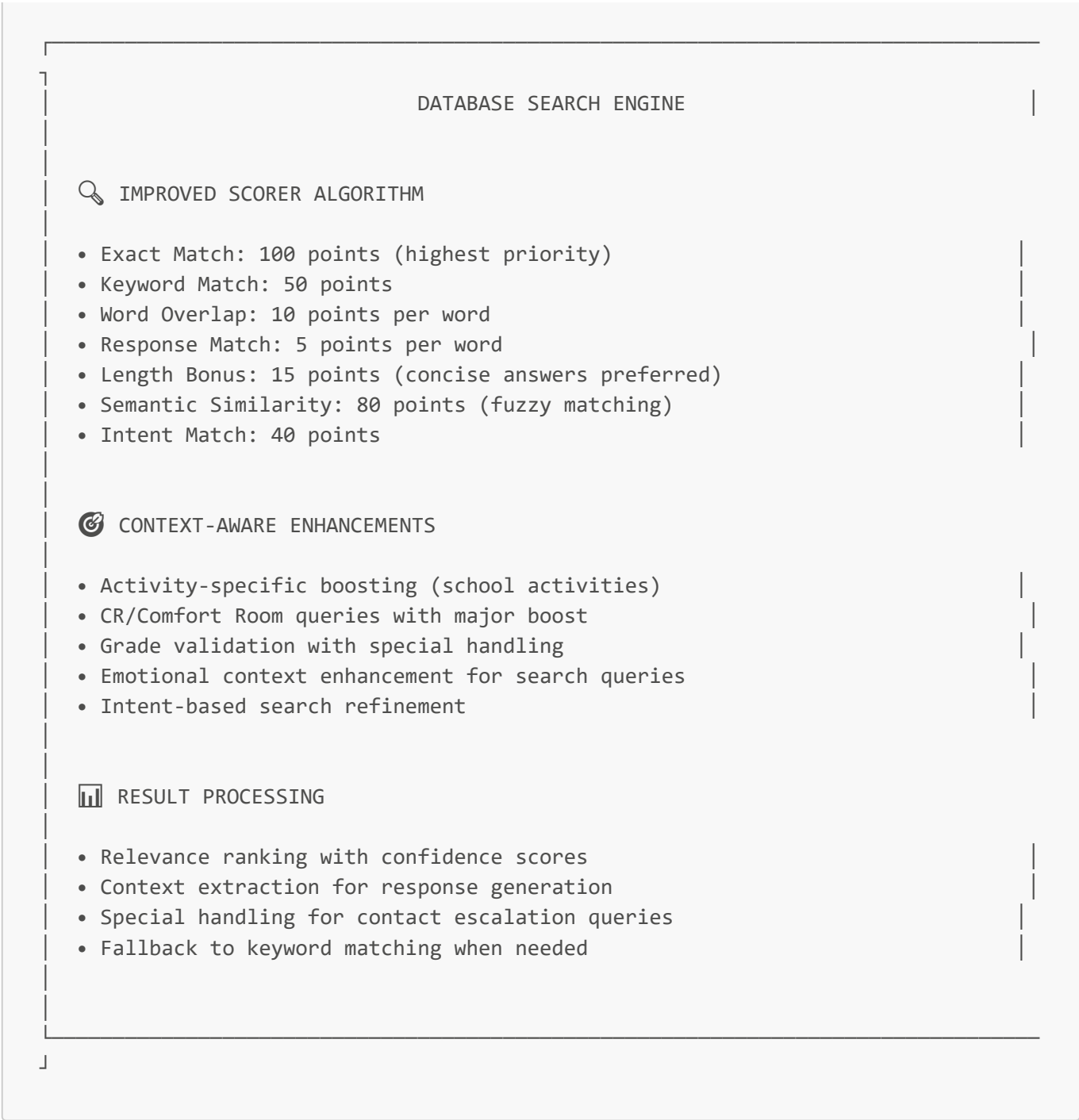
 ADVANCED AI FEATURES

- Emotional Intelligence with sentiment analysis
- Response Personalization based on user profiles
- Conversation Analysis with topic flow tracking
- Multi-provider AI with intelligent fallback
- Context-aware responses with memory integration
- Typo correction and text normalization
- Semantic similarity matching for database search

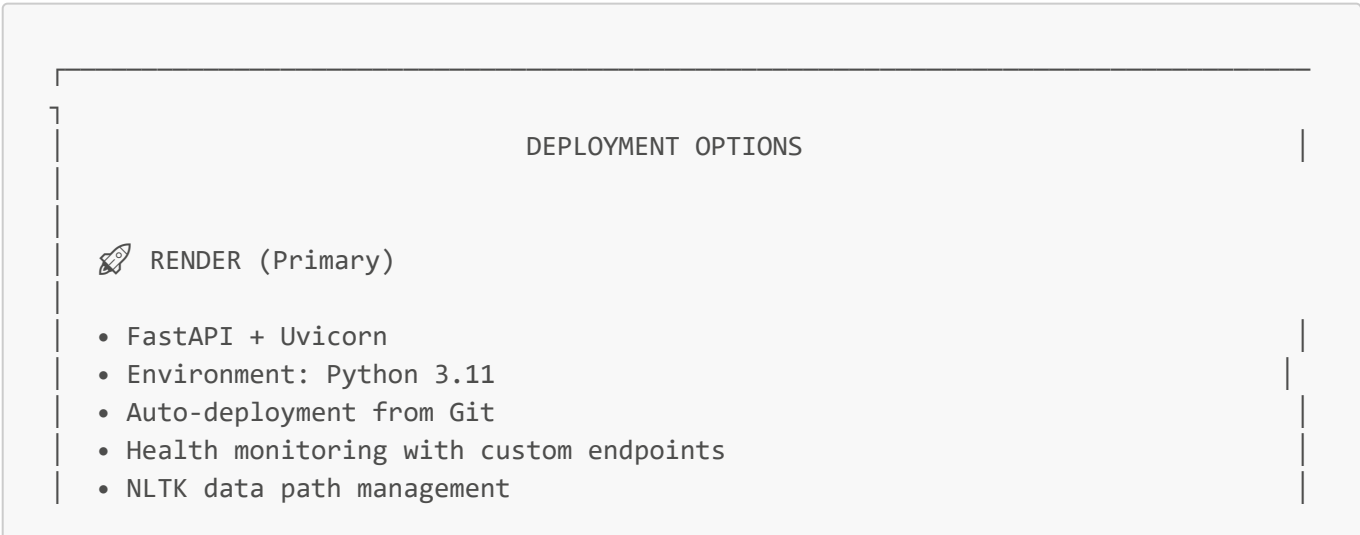
Data Flow (Updated)



Database Search System (Detailed)



Deployment Architecture (Updated)



 RAILWAY (Secondary)

- FastAPI + Gunicorn
- Environment: Python 3.11
- Railway.json configuration
- Health monitoring and metrics
- Optimized logging for production

 LOCAL DEVELOPMENT

- FastAPI + Uvicorn
- Environment: Python 3.11
- NLTK data in local directory
- Development logging with debug information

Security Features (Enhanced)

SECURITY LAYERS

 INPUT VALIDATION

- SQL Injection Protection Middleware
- XSS Prevention in responses
- Input Sanitization and validation
- Gibberish detection with NLP patterns

 RATE LIMITING

- Per-provider rate limits (Groq: 30/min, Cohere: 1000/min)
- Global rate limiting for abuse prevention
- Rate limit monitoring and alerts

 EMERGENCY DETECTION

- Medical emergency detection with context analysis
- Safety emergency detection for school incidents
- Humor/sarcasm detection to prevent false positives
- Standalone medical term validation
- Immediate 911 protocol activation

 DATA PROTECTION

- Encrypted API keys in environment variables
- Secure Supabase database connections
- Privacy-focused design with minimal data collection
- Session management with context clearing

Performance & Monitoring (Enhanced)

PERFORMANCE FEATURES

⚡ CACHING SYSTEM

- Language detection cache with 83% hit rate
- Translation cache for repeated phrases
- Response cache for common queries
- NLTK data caching for faster initialization

📊 MONITORING & METRICS

- Health endpoints (/health, /admin/logs)
- Performance metrics (/admin/metrics)
- Error tracking and logging
- Response time monitoring (<1.5s average)
- Cache hit rate monitoring
- Provider health status tracking

🔄 FALLBACK SYSTEM

- Multi-provider AI with intelligent switching
- Graceful degradation when providers fail
- Keyword matching fallback for critical queries
- Offline capabilities with cached responses
- Error recovery with user-friendly messages

🚀 OPTIMIZATION FEATURES

- Token-efficient system prompts
- Reduced logging for production environments
- Database timeout optimization (15s)
- Memory-efficient conversation handling
- Async processing for better performance

Key Innovations & Technical Achievements

TECHNICAL INNOVATIONS	
 MULTI-QUESTION PROCESSING • Detects 2-5 questions in single input • Structured response generation for each question • Context preservation across multiple questions • Intelligent question separation using NLP patterns	
 EMOTIONAL INTELLIGENCE • Real-time emotion detection (sad, happy, worried, etc.) • Sentiment analysis with confidence scoring • Empathy level assessment for response adaptation • Support needs identification	
 CONTEXT-AWARE TRANSLATION • Maintains conversation context during translation • Language-specific grammar preservation • Cultural context awareness • Confidence scoring for translation quality	
 ADVANCED DATABASE SEARCH • ImprovedScorer with weighted relevance scoring • Semantic similarity matching using difflib • Context-specific boosting for school topics • Intent-based search refinement	
 INTELLIGENT EMERGENCY DETECTION • Context-aware analysis to prevent false positives • Humor and sarcasm detection • Standalone medical term validation • Immediate safety protocol activation	

Intent Classifications (Complete List)

INTENT CLASSIFICATIONS	
 EMERGENCY INTENTS	
• EMERGENCY: Medical emergencies requiring immediate attention • MEDICAL_EMERGENCY: Specific medical emergencies • SAFETY_EMERGENCY: School safety emergencies	
 GREETING INTENTS	
• GREETING_SIMPLE: Basic greetings • GREETING_WITH_NAME: Greetings with name introduction • GREETING_RETURNING_USER: Greeting for returning users • GREETING_EXCITED: Enthusiastic greetings • GREETING_FORMAL: Formal/polite greetings • GREETING_CASUAL: Casual greetings	
 INTRODUCTION INTENTS	
• NAME_INTRODUCTION: User introduces themselves • CHILD_INTRODUCTION: User introduces their child • NAME_QUERY: User asking about their own name	
 SCHOOL INFORMATION INTENTS	
• SCHOOL_INFO: General school information • SCHOOL_OVERVIEW: School description/overview • GRADE_LEVELS: Questions about grades offered • SCHOOL_PROGRAMS: Academic programs and curriculum • FACILITIES_INQUIRY: Questions about school facilities • FINANCIAL_INQUIRY: Tuition, fees, payments	
 ENROLLMENT INTENTS	
• ENROLLMENT_INQUIRY: General enrollment questions • ENROLLMENT_DOCUMENTS: Document requirements • ENROLLMENT_DEADLINE: Enrollment deadlines • ENROLLMENT_PROCESS: Step-by-step enrollment process	

 CONTACT INTENTS

- STAFF_INQUIRY: Questions about teachers/staff
- CONTACT_INFO: Contact information requests
- CONTACT_ESCALATION: Requests to speak to a person

 CONVERSATION FLOW INTENTS

- FOLLOW_UP_QUESTION: Follow-up questions
- CLARIFICATION_REQUEST: Requests for more details
- TOPIC_CONTINUATION: Continuing previous topics
- COMPARISON_REQUEST: Comparing options/programs
- LOCATION_INQUIRY: School location questions
- SCHEDULE_INQUIRY: School hours/schedules

 EMOTIONAL/SOCIAL INTENTS

- EMOTIONAL_EXPRESSION: Expressing emotions
- APPRECIATION: Thank you messages
- HELP_REQUEST: General assistance requests
- CONFIRMATION: Yes/no responses
- DENIAL: User denying something
- CLARIFICATION: User clarifying meaning
- GOODBYE: User ending conversation

 FALLBACK INTENT

- UNKNOWN: When intent cannot be determined

Advanced NLP Capabilities

NLP CAPABILITIES

 INTENT CLASSIFICATION

- 40+ intent types with confidence scoring
- Rule-based and AI-assisted classification
- Multilingual pattern matching (English, Tagalog, Aklanon)
- Context-aware intent detection

 ENTITY EXTRACTION

- Named entity recognition (person names, locations, etc.)
- Relationship detection between entities
- Confidence scoring for extracted entities
- Context-aware entity extraction

 LANGUAGE DETECTION

- Multi-language support (English, Tagalog, Aklanon)
- Mixed-language input handling
- Context-aware language detection
- Confidence scoring for language detection

 CONVERSATION ANALYSIS

- Topic flow tracking across conversations
- Urgency level detection
- User expertise assessment
- Sentiment analysis and emotional state detection

 EMOTIONAL INTELLIGENCE

- Emotion detection (sad, happy, worried, etc.)
- Sentiment scoring and analysis
- Empathy level assessment
- Support needs identification

 RESPONSE PERSONALIZATION

- User profile creation and management
- Conversation history analysis
- Dynamic tone adjustment
- Formality level adaptation

 CONTEXT-AWARE TRANSLATION

- Maintains conversation context during translation
- Confidence scoring for translation quality
- Language-specific grammar handling
- Cultural context preservation

 MULTI-QUESTION PROCESSING

- Detects multiple questions in single input
- Parses and structures responses for each question

- Maintains context across multiple questions
- Handles 2-5 questions per input

 EMERGENCY DETECTION

- Medical emergency identification
- Safety protocol activation
- Context-aware analysis
- False positive reduction

 TEXT PREPROCESSING

- Typo correction and normalization
- Stop word removal and stemming
- Text cleaning and sanitization
- Character encoding handling

 SEMANTIC SIMILARITY

- Fuzzy matching using difflib
- Context-aware scoring
- Relevance ranking
- Similarity thresholds

 GIBBERISH DETECTION

- Pattern recognition for nonsensical input
- Entropy analysis for randomness detection
- Language-specific validation
- Character diversity analysis

This comprehensive architecture diagram represents a production-ready, multilingual chatbot system with advanced NLP capabilities, robust security measures, and intelligent fallback systems. It demonstrates sophisticated applied AI in education technology with comprehensive error handling, performance optimization, and real-world deployment considerations.